INCH-POUND

MIL-P-70674 (AR)
AMENDMENT 2
22 May 1991
SUPERSEDING
AMENDMENT 1
3 April 1989

MILITARY SPECIFICATION

PROJECTILE, 155MM: M864 METAL PARTS FOR BASE BURNER ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-70674 (AR), dated 4 April 1988, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title:

Change projectile nomenclature:

From: "XM864"

To: "M864"

1.1, Scope:

Change projectile nomenclature:

From: "XM864"

To: "M864"

* 2.1.1, Specifications and Standards:

Add the new military specification:

"MIL-STD-414 - Sampling Procedure and Table for Inspection by Variables for Percent Defective"

DISTIRBUTION STATEMENT A. Approved for public release; distribution is unlimited.

* 2.1.2, Other Government Documents, Drawings, and Publications:

Add the new Product Drawing (ARDEC):

"9381130 - Base Burner Assembly"

PAGE 2

* 2.2 Order of precedence:

Change paragraph:

From:

"2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

To:

"2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI Z1.1 - Guide for Quality Control Charts
ANSI Z1.2 - Control Chart Method of Analyzing Data
ANSI Z1.3 - Control Chart Method of Controlling Quality During Production

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, N.Y., N.Y. 10018.)"

* 2., APPLICABLE DOCUMENTS:

Add new subparagraph 2.3:

"2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

* 3.4.2, Process Control Dimensions:

Change the last sentence:

From: "Dimensions to be gaged for process control are to be assured to a .40% AQL."

To: "Dimensions to be gaged for process control are to be assured to a 0.40 AQL."

PAGE 6

* TABLE I:

Change item designation in title:

From: "XM864"

To: "M864"

PAGE 7

4.4.2, Examination:

Add new subparagraph 4.4.2.c:

" c. Statistical process control (SPC). For those dimensions identified in the Classification of Defects Tables by referencing Appendix A, the contractor must utilize the principles of SPC as defined herein, ANSI Z1.1, ANSI Z1.2, ANSI Z1.3 and Appendix A. When the contractor satisfies the control conditions identified in Appendix A, acceptance will then be by SPC. The SPC sample size and frequency of selection shall be determined by the contractor. If the SPC sample is to be less than four (4)

items per hour, then the approval of the procuring agency must be sought prior to the elimination of sampling inspection for acceptance. The contractor may request to implement SPC acceptance for any other feature that is not annotated in the classification for defects by requesting approval of the procuring agency and by meeting all the conditions specified in this specification. When not accepting product by SPC acceptance of product for those features identified by referencing Appendix A will only be in accordance with MIL-STD-414 or MIL-STD-1235 plans CSP - 1, -2,and -T using variable measurement capable gages."

PAGE 8

* 4.4.2.1, Cup:

Change paragraph by substituting new paragraph 4.4.2.1 (attached).

PAGE 9

* 4.4.2.2, Retainer:

Change paragraph by substituting new paragraph 4.4.2.2 (attached).

PAGE 10

* 4.4.2.3, Plug:

Change paragraph by substituting new paragraph 4.4.2.3 (attached).

PAGE 11

* 4.4.2.4, Housing, Igniter:

Change paragraph by substituting new paragraph 4.4.2.4 (attached).

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	8822518
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length	0.40	3.1/ App.A	Gage
102	Outside diameter	0.40	3.1/ App.A	Gage
Minor				
201	Inside diameter	0.65	3.1	Gage
202	Thickness thru bottom	0.65	3.1	Gage
203	Protective finish missing or incomplete	0.65	3.1	Visual
204	Evidence of poor workmanship	0.65	3.6	Visual

NOTES:

QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Retainer		9381110
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Length	0.40	3.1/ App.A	Gage
102	Thread pitch diameter	0.40	3.1/ App.A	Gage
103	Thread major diameter	0.40	3.1/ App.A	Gage
104	Largest inside diameter	0.40	3.1/ App.A	Gage
105	Smallest inside diameter	0.40	3.1/ App.A	Gage
106	Depth of largest inside diameter	0.40	3.1/ App.A	Gage
107	Largest outside diameter	0.40	3.1/ App.A	Gage
108	Deleted			
109	Length from front face to largest outside diameter	0.40	3.1	Gage
110	Depth of slots (2 places)	0.40	3.1	Gage
111	Width of slots	0.40	3.1	Gage
112	Length of thread pilot	0.40	3.1	Gage
113	Distance between slots	0.40	3.1	Gage
114	True position of slots	0.40	3.1	Gage
115	Length from rear face to shoulder	0.40	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Retainer	2 OF 2	9381110
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Width of thread relief	0.65	3.1	Gage
202	Diameter of thread pilot	0.65	3.1	Gage
203	Diameter of thread relief	0.65	3.1	Gage
204	Protective finish missing or incomplete	0.65	3.1	Visual
205	Evidence of poor workmanship	0.65	3.6	Visual

NOTES:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Plug		8597488
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Smallest inside diameter	0.40	3.1/ App.A	Gage
102	Largest inside diameter	0.40	3.1/ App.A	Gage
103	Pitch diameter	0.40	3.1/ App.A	Gage
104	Major diameter	0.40	3.1/ App.A	Gage
105	Distance between flats	0.40	3.1/ App.A	Gage
106	Depth of largest inside diameter	0.40	3.1/ App.A	Gage
107	Height of crown	0.40	3.1/ App.A	Gage
108	Length from bottom to top of cylindrical outside diameter	0.40	3.1/ App.A	Gage
Minor				
201	True position of smallest inside diameter	0.65	3.1	Gage
202	Cylindrical outside diameter	0.65	3.1	Gage
203	Length from bottom to bottom of flats (2 places)	0.65	3.1	Gage
204	Protective finish missing or incomplete	0.65	3.1	Gage
205	Evidence of poor workmanship	0.65	3.6	Visual

NOTES:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER 9381116
4.4.2.4	Housing, Igniter		NEXT HIGHER ASSEMBLY None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Base thread pitch diameter	0.40	3.1/ App.A	Gage
102	Base thread minor diameter	0.40	3.1/ App.A	Gage
103	Small thread pitch diameter (maximum)	0.40	3.1/ App.A	Gage
104	Small thread minor diameter (maximum)	0.40	3.1/ App.A	Gage
105	Total depth from front face	0.40	3.1/ App.A	Gage
106	Smallest inside diameter	0.40	3.1/ App.A	Gage
107	Depth from front face to end of base thread relief	0.40	3.1/ App.A	Gage
108	Thickness through closed end	0.40	3.1/ App.A	Gage
109	Depth from front face to start of smallest inside diameter	0.40	3.1/ App.A	Gage
110	Diameter of top row of port holes (random one hole)	0.40	3.1/ App.A	Gage

NOTES:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.4	Housing, Igniter		9381116
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
111	True position of slot	0.40	3.1/ App.A	Gage
112	True position of second largest inside diameter	0.40	3.1/ App.A	Gage
113	Any hole incomplete or blocked	0.40	3.1/ App.A	Gage
114	Diameter of middle row of port holes (random one hole)	0.40	3.1/ App.A	Gage
115	Diameter of bottom row of port holes (random one hole)	0.40	3.1/ App.A	Gage
Minor				
201	Total number of port holes	0.65	3.1	Visual
202	Location of port holes	0.65	3.1	Gage
203	Second smallest outside diameter	0.65	3.1	Gage
204	Second largest outside diameter	0.65	3.1	Gage
205	Depth of base thread pilot	0.65	3.1	Gage
206	Diameter of base thread pilot	0.65	3.1	Gage
207	Total length	0.65	3.1	Gage
208	Width of base thread pilot	0.65	3.1	Gage
209	Smallest outside diameter	0.65	3.1	Gage
210	Largest outside diameter	0.65	3.1	Gage

NOTES:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Housing, Igniter	3 OF 3	9381116
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
211	Diameter of base thread relief	Ø.65	3.1	Gage
212	Second largest inside diameter	Ø.65	3.1	Gage
213	Length of largest outside diameter	Ø.65	3.1	Gage
214	Deleted			
215	Length of smallest outside diameter	Ø.65	3.1	Gage
216	Small thread relief diameter	Ø.65	3.1	Gage
217	Depth from front face to rear of second largest inside diameter	Ø.65	3.1	Gage
218	Width of small thread relief	Ø.65	3.1	Gage
219	Protective finish missing or incomplete	Ø.65	3.1	Gage
220	Evidence of poor workmanship	Ø.65	3.6	Visual
221	Small thread pitch diameter, minimum, after protective finish	Ø.65	3.1	Gage

NOTES:

* 5.2, Domestic Shipment Retainer and Igniter Housing:

Change:

From:

"5.2 Domestic shipment of retainer and igniter housing. Packaging, packing and marking for the retainer and for the igniter housing shall be in accordance with MIL-STD-1169, Level B. In addition, packaging shall be, as a minimum, single item packaging as stated in the paragraph entitled "Arrangements" of MIL-STD-1169 and moisture proof as stated in the paragraph entitled "Item Protection" of MIL-STD-1169. Additional packaging shall be provided as required to prevent wearing of the oxide coating during shipment."

To:

5.2 Domestic shipment of retainer and igniter housing. Packaging, packing and marking for the igniter housing shall be in accordance with MIL-STD-1169, Level B. In addition, packaging shall be, as a minimum, single item and moistureproof using heat sealed barrier bags in accordance with the Level B requirements of MIL-STD-1169. Additional packaging shall be provided, as required, to prevent wearing of phosphate coating."

* 6. Notes:

Change:

From:

"6. NOTES"

To:

"6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1, Intended use.

Change Projectile Nomenclature:

From: "XM864"

To: "M864"

* 6.2 Ordering Data:

Change:

From:

"6.2 Ordering data. See MIL-A-48078."

To:

"6.2 Acquisition requirements. See MIL-A-48078."

* 6.3 Submission of inspection equipment designs for approval:

Change:

From:

"See MIL-A-48078. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Dover, NJ 07801- 5001."

To:

"See MIL-A-48078. Submit designs as required to: Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

* Add new paragraph 6.10:

"6.10 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability."

* 6.8, Subject term (keyword) listing:

Change Projectile Nomenclature

From: Projectile, 155MM, XM864, Metal Parts For

To: Projectile, 155MM, M864, Metal Parts For

Append the attached Appendix A to the specification.

APPENDIX A

10. SCOPE

10.1 Scope. This appendix details the procedure for implementation of statistical process control for articles covered in this specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Application of statistical process control (SPC). The contractor will, for the identified dimensions, maintain Xbar and R SPC charts. After the contractor has demonstrated that his process for manufacturing a particular feature is in control, as defined in 30.1.1; he will discontinue acceptance sampling inspection in favor of acceptance by SPC. However, SPC charting will be maintained for all identified dimensions whether or not the process is in control. It is expected that the contractor will make every effort to bring each process into control. It is required that the contractor re-evaluate his established control limits on a monthly basis and make those changes warranted by the data. At no time should the process parameters vary such that it will fail to meet the requirements established in subsequent paragraphs. Should this happen, the contractor will return to acceptance by sampling.

30.1.1 Implementation of SPC for acceptance of product. For those dimension identified in the classifications of defects tables as being applicable for elimination of lot sampling acceptance, the contractor shall implement, on a dimension by dimension basis, acceptance by SPC after he conforms to each of the following steps:

a. The contractor performs an in depth process capability study to establish the mean process average (Xdouble bar), Rbar, upper and lower control limits. This study will consist of at least 40 subgroups of data collection.

b. The contractor demonstrates that his process is under control for at least 20 consecutive production days. Control is defined as remaining within the established control parameters and not exhibiting any of the out of control conditions defined in 30.1.2.a thru 30.1.2.e more than two times on an individual character basis.

APPENDIX A

c. Thirty days consecutive production have passed acceptance sampling for the feature.

d. The process has a demonstrated Cpk equal to or greater than 1.3 as demonstrated when conducting the above test. Cpk is defined as follows:

$$Cpk = \frac{\text{process mean - nearest specification limit}}{3 \text{ sigma}}$$

sigma = process standard deviation (calculated by the range method per the referenced ANSI documents

e. The contractor documents his detailed process description. This process description must be submitted to Commander, AMCCOM, ATTN: AMSMC-QAR-Q(D), Picatinny Arsenal, New Jersey 07806-5000 for review and disapproval if the process description does not meet its objectives. That is that the process description will identify all known contributing variables and their respective controls that are necessary to maintain the specific dimension (i.e., weight, temperature, machine speed, time, tool life, solution proportion, etc.). The Government will furnish the contractor with a written notice of the acceptability of his process controls. A notice of control acceptability is required prior to the implementation of acceptance by SPC.

Process control analysis. Once SPC acceptance is implemented, the contractor will evaluate his process for evidence of non-random variation (e.g. trends, extreme values, etc.). Should any out of control condition be detected, the product produced from the last sample until the process is brought back into control will be isolated for analysis. The contractor will bring the process back into control or shut the process down. Upon determining the cause of the out of control conditions, the contractor will determine if the product manufactured and isolated could be out of engineering tolerance. If the out of control condition did not result in the production of nonconforming material, the material will be released. If the out of control condition could have created nonconforming product, the contractor will sample the product using MIL-STD-105 and the AQL specified in the classification of defects table to determine the acceptability of the isolated product. This evaluation of out of control product requires the review and approval of the local Government quality assurance representatives. As a minimum, the contractor will look for any of the following out of control conditions:

a. If any point goes outside the Xbar or R chart control limits.

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b. If 9 points in a row are either continually above or continually below the Xdouble bar.

c. If 6 points in a row are either continually increasing or continually decreasing on the Xbar chart.

d. If 6 points in a row are increasing on the range chart.

e. If any individual sample, Xi, falls outside of the specification limits.

30.1.2.1 Process capability review. At least once a month the contractor will review his process data and establish updated control limits.

30.1.3 Mandatory return to sampling acceptance. Acceptance by SPC will be discontinued in accordance with the following paragraphs. In any case, control charting will be continued. When this occurs, the contractor may reintroduce acceptance by SPC by repeating the steps outlined in paragraph 30.1.1.

30.1.3.1 Out of control incidents. If for any individual process, an out of control incident is identified 3 times within 5 consecutive shifts or 5 times within 30 consecutive shifts; and the analysis indicated that nonconforming product may have been manufactured, the contractor will return to acceptance by sampling the pertinent feature.

30.1.3.2 Process interruption. If the process is interrupted for more than 10 calendar days, the contractor will return to acceptance by sampling for that pertinent feature.

30.1.3.3 Loss of control. If either by testing or data review it is determined that the Cpk is no longer at the required minimum of 1.3, the contractor will return to acceptance by sampling for the pertinent feature.

30.1.4 Records. The contractor will maintain all records dealing with SPC in his files for a minimum of 5 years after completion of the contract.

30.2 Government auditing. In order to verify the state of control of the process, it is recommended that a Government representative select samples at unscheduled times. The sample mean and range should then be compared to the data being recorded by the contractor. If the data is not consistent with the reported data, corrective action should be sought and, if necessary, screening of material should be required.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-AC02)